

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)

QUESTIONS: 5

Total Marks:50

Annexure A

Experiments

Criteria	Understanding of Concepts (2.5)	Experimental Design (3)	Use of Tools (2)	Analysis of Results (1.5)	Documentation (1)	Total(10)
Weightage	25%	30%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I P.Manoj, 192411081 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

ANNEXURE C

Experiments :

1. Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Photograph:

Age of child	A	B	C
5-6 years	18	22	20
7-8 years	2	28	40
9-10 years	20	10	40

Use `cov()` to calculate the sample covariance between B and C.

Use another call to `cov()` to calculate the sample covariance matrix for the preferences.

Use `cor()` to calculate the sample correlation between B and C.

Use another call to `cor()` to calculate the sample correlation matrix for the preferences.

Answer:

Step 1 – Chi-square test of independence (Age vs Photograph preference)

Row totals: 5-6 yrs = 60, 7-8 yrs = 70, 9-10 yrs = 70. Column totals: A = 40, B = 60, C = 100. Grand total = 200.

Expected count = (row total × column total) / grand total, giving the table:

Age of child	A (Exp)	B (Exp)	C (Exp)
5-6 years	12	18	30
7-8 years	14	21	35
9-10 years	14	21	35

$$\chi^2 = \Sigma(\text{O}-\text{E})^2/\text{E} \approx 7.22 + 13.33 + 9.05 = 29.60$$

Degrees of freedom = (rows-1)×(cols-1) = 2×2 = 4. Critical value ($\alpha = 0.05$, df = 4) = 9.49.

Since 29.60 > 9.49, we reject the null hypothesis — there IS a statistically significant relationship between a child's age and photograph preference.

What is wrong with this study?

- Each child is forced to pick exactly one photo, so a genuine “no preference” response is impossible.
- The photos themselves may differ in attractiveness, familiarity, or expression — the study does not control for what specifically drives the preference, so the effect may reflect photo content rather than age.
- Sampling is not described as random and group sizes/ages are coarse bands, which can mask real developmental patterns.
- A significant chi-square result shows association, not causation — it does not explain WHY age relates to preference.

R computations:

```
A <- c(18, 2, 20); B <- c(22, 28, 10); C <- c(20, 40, 40)
prefs <- data.frame(A, B, C)
cov(B, C)           # sample covariance between B and C
cov(prefs)          # sample covariance matrix
cor(B, C)           # sample correlation between B and C
cor(prefs)          # sample correlation matrix
```

Results:

`cov(B, C)` = -20.0

Covariance matrix $\approx$ [A: 97.33, -74.00, -46.67] / [B: -74.00, 84.00, -20.00] / [C: -46.67, -20.00, 133.33]

$\text{cor}(B, C) \approx -0.189$ (weak negative correlation)

Correlation matrix $\approx$ [A: 1.00, -0.819, -0.410] / [B: -0.819, 1.00, -0.189] / [C: -0.410, -0.189, 1.00]

Interpretation: A and B are strongly negatively correlated (as A rises, B falls), while B and C are only weakly related.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID	Player Name	Points Scored
P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17
P13	Player M	19
P14	Player N	21
P15	Player O	23

Answer:

Sorted points ($n = 15$): 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 35

Median = 8th value = 17

Q1 (median of lower 7: 10,11,12,13,14,15,16) = 13

Q3 (median of upper 7: 18,19,20,21,22,23,35) = 21

IQR = $Q3 - Q1 = 21 - 13 = 8$

Lower fence = $Q1 - 1.5 \times \text{IQR} = 13 - 12 = 1$ Upper fence = $Q3 + 1.5 \times \text{IQR} = 21 + 12 = 33$

Any value outside [1, 33] is an outlier. Player J (P10) scored 35, which is above the upper fence (33) — so it is flagged as a scoring outlier.

Boxplot construction: draw a box from Q1 (13) to Q3 (21) with a line at the median (17); whiskers extend to the smallest/largest points within the fences (10 to 23); Player J's score of 35 is plotted as an individual point beyond the upper whisker, visually marking it as an outlier.

3. A bank wants to decide whether to approve a loan based on customer attributes like: Age, Income, Employment Status, Credit Score. Using historical data, build a Decision Tree classifier to predict Loan Approval (Yes/No). (10 Marks)

Consider the Data Set (15 records, 9 'yes' / 6 'no') as given in the question.

Answer:

Root entropy: $p(\text{yes})=9/15=0.6$, $p(\text{no})=6/15=0.4 \rightarrow \text{Entropy}(S) = -0.6\log_2(0.6) - 0.4\log_2(0.4) = 0.971$

Splitting on Credit Score:

- poor (5 records) $\rightarrow$ all 5 are 'No', entropy = 0
- good (5 records) $\rightarrow$ all 5 are 'Yes', entropy = 0
- average (5 records) $\rightarrow$ 4 Yes, 1 No, entropy = 0.722

Weighted entropy = $(5/15) \times 0 + (5/15) \times 0 + (5/15) \times 0.722 = 0.241 \rightarrow \text{Information Gain}(\text{Credit Score}) = 0.971 - 0.241 = 0.730$

This is far higher than the information gain from Age (≈ 0.083), Income, or Employment Status, so Credit Score is chosen as the root node.

For the mixed 'average' branch, splitting further on Income cleanly separates the remaining cases (Income=low $\rightarrow$ No; Income=medium/high $\rightarrow$ Yes), matching the single 'No' record (old, low income, self-employed, average credit).

Resulting Decision Tree:

Credit Score = poor $\rightarrow$ No

Credit Score = good $\rightarrow$ Yes

Credit Score = average:

Income = low $\rightarrow$ No

Income = medium or high $\rightarrow$ Yes

This tree classifies all 15 training records correctly and shows that Credit Score is the strongest predictor of loan approval, with Income refining the decision only for 'average' credit applicants.

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 50

(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences

Answer:

Class A sorted: 35, 36, 47, 64, 66, 76, 84, 89, 95 (n=9)

Mean = $592/9 = 65.78$ Median = 66 Range = $95 - 35 = 60$

Q1 = 41.5, Q3 = 86.5 (IQR = 45)

Class B sorted: 50, 51, 56, 59, 60, 63, 66, 70, 84 (n=9)

Mean = $559/9 = 62.11$ Median = 60 Range = $84 - 50 = 34$

Q1 = 53.5, Q3 = 68 (IQR = 14.5)

(i) Class A scored the higher mean (65.78 vs 62.11) and higher median (66 vs 60), but also the larger range (60 vs 34).

(ii) Boxplot inference: Class A's box and whiskers are noticeably wider (IQR = 45) than Class B's (IQR = 14.5), showing Class A's results are more spread out / inconsistent, with scores ranging from a low of 35 to a high of 95. Class B's boxplot is tighter and more compact, showing more consistent performance clustered around the low-to-mid 50s–70s, though its top score (84) stretches toward its upper whisker. Overall: Class A performed better on average but with much greater variability between students; Class B was more uniform but scored somewhat lower overall. Neither class shows a statistical outlier by the $1.5 \times \text{IQR}$ rule.

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2

age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median, and standard deviation of age and %fat.

(b) Draw the boxplots for age and %fat.

(c) Draw a scatter plot and a q-q plot based on these two variables.

Answer:

(a) Descriptive statistics (n = 18):

age: Mean = 46.44, Median = 51, Standard deviation $\approx$ 13.22

%fat: Mean = 28.78, Median = 30.7, Standard deviation $\approx$ 9.26

(b) Boxplots:

age — Q1 = 39, Median = 51, Q3 = 57 (IQR = 18); fences = [12, 84]; all 18 ages fall inside the fences, so the age boxplot shows no outliers, with a moderately wide, fairly symmetric spread from 23 to 61.

%fat — Q1 = 26.5, Median = 30.7, Q3 = 34.1 (IQR = 7.6); fences = [15.1, 45.5]. The two lowest readings, 7.8% and 9.5%, fall below the lower fence of 15.1 and are flagged as low outliers on the %fat boxplot — these correspond to the two 23- and 27-year-old subjects with unusually low body fat. The rest of the box is fairly narrow and symmetric around the median.

(c) Scatter plot & Q-Q plot:

Scatter plot (age on x-axis, %fat on y-axis): the points show a moderate positive trend — %fat generally increases as age increases, though the relationship is not perfectly linear and the two low-%fat points at young ages stand out as departures from the trend.

Q-Q plot: plotting the sample quantiles of age (and separately of %fat) against theoretical normal quantiles shows age tracking the reference line reasonably well across the middle of the distribution (roughly normal), while the %fat Q-Q plot bends away from the line at the lower tail because of the two low outliers (7.8, 9.5), indicating %fat is not perfectly normally distributed at the low end.

R code:

```
age <- c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)
fat <- c(9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,34.6,42.5,28.8,33.4,30.2,34.1,32.9,41.2,35.7)
mean(age); median(age); sd(age)
mean(fat); median(fat); sd(fat)
boxplot(age, main="Age"); boxplot(fat, main="%Fat")
plot(age, fat, main="Age vs %Fat")
qqnorm(age); qqline(age)
qqnorm(fat); qqline(fat)
```